

NETWORKS AND COMPLEXITY

Solution 15-9

*This is an example solution from the forthcoming book Networks and Complexity.
Find more exercises at <https://github.com/NC-Book/NCB>*

Ex 15.9: aSIS invasion threshold done differently [Lvl. 4]

Consider a typical newly infected node in the aSIS model in the scenario where the disease is still very rare. Find a way to compute the secondary infections R_0 that this node causes before it recovers. Does this give you the same threshold for disease invasion as above? (Because other susceptible nodes will rewire their links away from your focal node you may need to integrate in time. As the threshold will happen at very small infection rates you can use that p is small)

Solution

Let's define x as the probability that the node is still infected at time t after infection. We know that this probability decreases at rate r , so

$$\dot{x} = -rx \quad (1)$$

with initial condition $x(0) = 1$ and hence

$$x(t) = e^{-rt} \quad (2)$$

Since the infection reaches our node via a link we can assume that we have $q = z$ additional neighbors. And, because the disease is rare these additional neighbors are initially susceptible. Since our focal node is infected the susceptible neighbors will start to rewire their links away at rate w . Hence the number of susceptible neighbors, k will change according to

$$\dot{k} = -wk \quad (3)$$

with initial condition $k(0) = z$ which we integrate to find

$$k(t) = ze^{-wt} \quad (4)$$

The rate at which we infect neighbors as long as we are infected is p . So if we multiply this with the number of neighbors and the probability that we are still infected we get the instantaneous rate of transmissions $pze^{-rt}e^{-wt} = pze^{-(r+w)t}$. To find our R_0 we integrate over all time

$$R_0 = \int pze^{-(r+w)t} dt = \left[-\frac{pz}{r+w} e^{-(r+w)t} \right]_0^\infty = 0 + \frac{pz}{r+w} \quad (5)$$

and hence

$$R_0 = \frac{pz}{r+w} \quad (6)$$

Our disease can spread if $R_0 > 1$ which we can write as

$$pz > r + w \quad (7)$$

which is the same that we found above in a different way.